SPECTRA

2026 • HSTU, DINAJPUR, BANGLADESH

CALL FOR PAPERS

Symposium on Photonics, Emerging Computational Technologies, Research and AI Data Science

CALL FOR PAPERS

REGISTRATION DEADLINE
31 April 2026

ACCEPTANCE NOTIFICATION
3 May 2026

CONFERENCE DATE
15-16 May 2026

OVERVIEW

SPECTRA 2026 invites submissions at the intersection of photonics and artificial intelligence. Topics of interest include AI-driven photonic device design, optical neural networks, intelligent sensing, photonics for AI hardware acceleration, and machine learning for imaging and spectroscopy. Accepted papers will be presented at the symposium.

View Details

https://ieeespectra.tech

Submission URL

forms.gle/y7zz6sXuADebW6NV9

SCAN HERE

SYMPOSIUM SCOPES

submit your research ideas under any of the following tracks

PHOTONICS & OPTICAL SCIENCE

- Optical Communications & Networks
- Integrated & Silicon Photonics
- Fiber Optics & Photonic Waveguides
- Quantum Photonics & Quantum Information
- Photonic Sensors & Instrumentation
- Imaging & Optical Measurement Systems
- Photonic Materials & Fabrication

EMERGING COMPUTATIONAL TECHNOLOGY

- High-Performance Computing (HPC)
- Edge, Fog & Neuromorphic Computing
- Photonic Computing & Optical Processors
- Embedded Systems & VLSI Design
- Internet of Things (IoT) & IIoT
- Microcontroller-Based Applications
- System-on-Chip & Hardware Acceleration

ARTIFICIAL INTELLIGENCE & DATA SCIENCE

- Machine Learning & Deep Learning
- Data Engineering, Mining & Analytics
- Computer Vision & Perception
- Natural Language & Speech Processing
- AI in Healthcare & Biomedical Apps
- AI in Decision Support Systems
- Human-AI Interaction & Explainable AI

AGRICULTURE, FOOD & ENVIRONMENT

- Smart Agriculture & Precision Farming
- Agricultural Data Science & Remote Sensing
- Climate & Environmental Informatics
- Biosensing & Food Safety Technologies
- Fisheries & Aquaculture Monitoring
- Veterinary Biomedical Technologies

NATURAL SCIENCE ENABLING TECHNOLOGY

- Photonic & Semiconductor Physics
- Material Science for Optoelectronics
- Spectral Analysis & Chemical Sensing
- Computational Physics & Chemistry
- Simulation, Modeling & Optimization
- Experimental Instrumentation & Lab Innovations

HEALTHCARE, MEDICINES & LIFE SCIENCE

- Biomedical Signal & Image Analysis
- Bioinformatics & Computational Biology
- Diagnostic Instrumentation & Wearables
- Assistive & Rehabilitation Technologies

CROSS DISCIPLINARY & FUTURE TECHNOLOGY

- AI-Photonics Convergence
- Computing for Physical Sciences
- Smart Infrastructure & Industry 4.0
- Sustainable Energy & Green Technology
- National Digital Transformation & Smart Governance
- Climate Technology & Disaster Preparedness
- Economic Development Through Technology

BUSINESS, ECONOMICS & SOCIAL SYSTEMS

- Business Analytics & Data-Driven Decision
- Technology Management & Innovation Strategy
- FinTech, InsurTech & Digital Commerce
- Tech-enabled Supply Chain & Logistics
- ICT for Social Sciences & Public Policy

EDUCATION, HUMANITIES & COMMUNICATION

- Digital Humanities & Computational Linguistics
- EdTech & Learning Analytics
- Technology Adoption & UX
- Communication in Scientific Dissemination
- Ethics, Policy & Societal Impacts of AI

HOSTED BY
Department of ECE

ORGANIZED BY
IEEE Photonics Society

CO-ORGANIZED BY
ECE Club of HSTU